

LACAITA ANDREA LEONARDO

professor

基本信息

姓名: LACAITA ANDREA LEONARDO 所属院校: 米兰理工大学

地区: Italy 年龄范围: 50岁以上

荣誉称号

★ IEEE Fellow

这些成就代表了在学术领域的最高认可, 展现了卓越的研究能力和影响力。

学科领域

微电子科学与工程

集成电路设计与集成系统

研究领域

微电子器件与制造

纳米电子与先进工艺

射频集成电路

模拟与数模混合电路

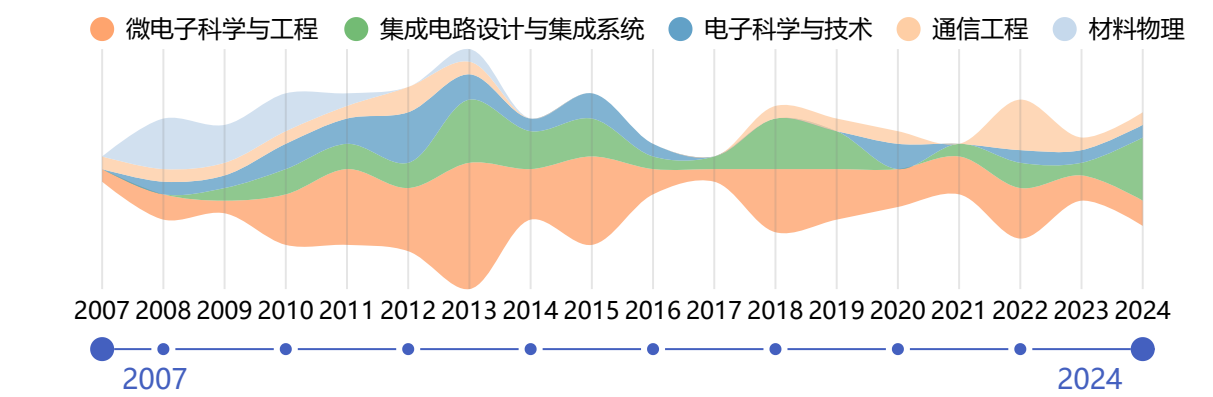
存储器设计与架构

个人简介

Prof. LACAITA ANDREA LEONARDO is a Full Professor of Electronics at Politecnico di Milano, teaching "Electron Devices " and "Analog Circuit Design" . He received his "Laurea" in Nuclear Engineering in 1985. He was a Researcher at CNR from 1987 - 1992, became an Associate Professor in 1992 and a Full Professor in 2000. His research focuses on electronics, especially electronic circuit design. He was elected IEEE Fellow in 2009. He has been a Visiting Scientist at several institutions, served on multiple conference

committees, and held various leadership positions. He has co - authored over 350 papers and a book. His office is in Building 22, Floor 4, Office 006, and his phone number is 6117.

近年发表的论文



A 59.3fs Jitter and-62.1dBc Fractional-Spur Digital PLL Based on a Multi-Edge Power-Gating Phase-Detector
2024 IEEE CUSTOM INTEGRATED CIRCUITS CONFERENCE, CICC
Dartizio, Simone M.; Rossoni, Michele; Tesolin, Francesco; Castoro, Giacomo; Samori, Carlo; Lacaita, Andrea L.; Levantino, Salvatore
被引用次数: 1 2024

A 66.7fs-Integrated-Jitter Fractional-N Digital PLL Based on a Resistive-Inverse-Constant-Slope DTC
2024 IEEE CUSTOM INTEGRATED CIRCUITS CONFERENCE, CICC
Salvi, Pietro; Dartizio, Simone M.; Rossoni, Michele; Tesolin, Francesco; Castoro, Giacomo; Lacaita, Andrea L.; Levantino, Salvatore
被引用次数: 4 2024

A Low-Noise Fractional-N Digital PLL Using a Resistor-Based Inverse-Constant-Slope DTC
IEEE JOURNAL OF SOLID-STATE CIRCUITS
Salvi, Pietro; Dartizio, Simone M.; Rossoni, Michele; Tesolin, Francesco; Castoro, Giacomo; Lacaita, Andrea L.; Levantino, Salvatore
被引用次数: 1 2024

A Low-Jitter Fractional-N Digital PLL Adopting a Reverse-Concavity Variable-Slope DTC
IEEE JOURNAL OF SOLID-STATE CIRCUITS
Rossoni, Michele; Dartizio, Simone M.; Tesolin, Francesco; Castoro, Giacomo; Dell'Orto, Riccardo; Lacaita, Andrea L.; Levantino, Salvatore
被引用次数: 0 2024

A 10-GHz Digital-PLL-Based Chirp Generator With Parabolic Non-Uniform Digital Predistortion for FMCW Radars

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Tesolin, Francesco; Dartizio, Simone M.; Castoro, Giacomo; Buccoleri, Francesco; Rossoni, Michele; Cherniak, Dmytro; Samori, Carlo; Lacaita, Andrea L.; Levantino, Salvatore

被引用次数: 0

2024

A 2-GS/s Time-Interleaved ADC With Embedded Background Calibrations and a Novel Reference Buffer for Reduced Inter-Channel Crosstalk

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Ricci, Luca; Be, Gabriele; Rocco, Michele; Scaletti, Lorenzo; Zanoletti, Gabriele; Bertulesi, Luca; Lacaita, Andrea L.; Levantino, Salvatore; Samori, Carlo; Bonfanti, Andrea

被引用次数: 0

2024

A Low-Spur and Low-Jitter Fractional-N Digital PLL Based on an Inverse-Constant-Slope DTC and FCW Subtractive Dithering

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Dartizio, Simone M.; Tesolin, Francesco; Castoro, Giacomo; Buccoleri, Francesco; Rossoni, Michele; Cherniak, Dmytro; Samori, Carlo; Lacaita, Andrea L.; Levantino, Salvatore

被引用次数: 10

2023

A Novel LO Phase-Shifting System Based on Digital Bang-Bang PLLs With Background Phase-Offset Correction for Integrated Phased Arrays

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Tesolin, Francesco; Dartizio, Simone M.; Buccoleri, Francesco; Santiccioli, Alessio; Bertulesi, Luca; Samori, Carlo; Lacaita, Andrea L.; Levantino, Salvatore

被引用次数: 4

2023

Phase Noise Analysis of Periodically ON/OFF Switched Oscillators

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Castoro, Giacomo; Dartizio, Simone M.; Lacaita, Andrea L.; Levantino, Salvatore

被引用次数: 3

2023

A Fractional-N Bang-Bang PLL Based on Type-II Gear Shifting and Adaptive Frequency Switching Achieving 68.6 fs-rms-Total-Integrated-Jitter and 1.56 μ s-Locking-Time

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Dartizio, Simone M.; Buccoleri, Francesco; Tesolin, Francesco; Avallone, Luca; Santiccioli, Alessio; Iesurum, Agata; Steffan, Giovanni; Cherniak, Dmytro; Bertulesi, Luca; Bevilacqua, Andrea; Samori, Carlo; Lacaita, Andrea L.; Levantino, Salvatore

被引用次数: 14

2022

Novel Feed-Forward Technique for Digital Bang-Bang PLL to Achieve Fast Lock and Low Phase Noise

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Bertulesi, Luca; Cherniak, Dmytro; Mercandelli, Mario; Samori, Carlo; Lacaita, Andrea L.; Levantino, Salvatore

被引用次数: 7

2022

A 12.9-to-15.1-GHz Digital PLL Based on a Bang-Bang Phase Detector With Adaptively Optimized Noise Shaping

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Dartizio, Simone M.; Tesolin, Francesco; Mercandelli, Mario; Santiccioli, Alessio; Shehata, Abanob; Karman, Saleh; Bertulessi, Luca; Buccoleri, Francesco; Avallone, Luca; Parisi, Angelo; Lacaita, Andrea L.; Kennedy, Michael P.; Samori, Carlo; Levantino, Salvatore

被引用次数: 16

2022

A 12.5-GHz Fractional-N Type-I Sampling PLL Achieving 58-fs Integrated Jitter

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Mercandelli, Mario; Santiccioli, Alessio; Parisi, Angelo; Bertulessi, Luca; Cherniak, Dmytro; Lacaita, Andrea L.; Samori, Carlo; Levantino, Salvatore

被引用次数: 28

2022

Analysis and Design of 8-to-101.6-GHz Injection-Locked Frequency Divider by Five With Concurrent Dual-Path Multi-Injection Topology

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Garghetti, Alessandro; Lacaita, Andrea L.; Seebacher, David; Bassi, Matteo; Levantino, Salvatore

被引用次数: 4

2022

A 900-MS/s SAR-Based Time-Interleaved ADC With a Fully Programmable Interleaving Factor and On-Chip Scalable Background Calibrations

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Be, Gabriele; Parisi, Angelo; Bertulessi, Luca; Ricci, Luca; Scaletti, Lorenzo; Mercandelli, Mario; Lacaita, Andrea L.; Levantino, Salvatore; Samori, Carlo; Bonfanti, Andrea

被引用次数: 7

2022

A Generalization of the Groszkowski's Result in Differential Oscillator Topologies

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Buccoleri, Francesco; Bonfanti, Andrea; Lacaita, Andrea L.

被引用次数: 3

2021

A PLL-Based Digital Technique for Orthogonal Correction of ADC Non-Linearity

2021 28TH IEEE INTERNATIONAL CONFERENCE ON ELECTRONICS, CIRCUITS, AND SYSTEMS (IEEE ICECS 2021)

Parisi, Angelo; Mercandelli, Mario; Samori, Carlo; Lacaita, Andrea L.

被引用次数: 0

2021

Random Telegraph Noise in 3D NAND Flash Memories

MICROMACHINES

Spinelli, Alessandro S.; Malavena, Gerardo; Lacaita, Andrea L.; Monzio Compagnoni, Christian

被引用次数: 8

2021

Characterization and Modeling of Current Transport in Metal/Ferroelectric/Semiconductor Tunnel Junctions

IEEE TRANSACTIONS ON ELECTRON DEVICES

Franchini, Giulio; Spinelli, Alessandro S.; Nicosia, Gianluca; Fumagalli, Ivan; Asa, Marco; Groppi, Chiara; Rinaldi, Christian; Lacaita, Andrea L.; Bertacco, Riccardo; Monzio Compagnoni, Christian

被引用次数: 10

2020

Variability Effects in Nanowire and Macaroni MOSFETs-Part I: Random Dopant Fluctuations

IEEE TRANSACTIONS ON ELECTRON DEVICES

Spinelli, Alessandro S.; Compagnoni, Christian Monzio; Lacaita, Andrea L.

被引用次数: 16

2020

Variability Effects in Nanowire and Macaroni MOSFETs-Part II: Random Telegraph Noise

IEEE TRANSACTIONS ON ELECTRON DEVICES

Spinelli, Alessandro S.; Compagnoni, Christian; Lacaita, Andrea L.

被引用次数: 7

2020

A 66-fs-rms Jitter 12.8-to-15.2-GHz Fractional-N Bang-Bang PLL With Digital Frequency-Error Recovery for Fast Locking

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Santiccioli, Alessio; Mercandelli, Mario; Bertulessi, Luca; Parisi, Angelo; Cherniak, Dmytro; Lacaita, Andrea L.; Samori, Carlo; Levantino, Salvatore

被引用次数: 55

2020

Time-Variant Modeling and Analysis of Multiplying Delay-Locked Loops

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Santiccioli, Alessio; Samori, Carlo; Lacaita, Andrea L.; Levantino, Salvatore

被引用次数: 15

2019

Compact modeling of GIDL-assisted erase in 3-D NAND Flash strings

JOURNAL OF COMPUTATIONAL ELECTRONICS

Malavena, Gerardo; Mannara, Aurelio; Lacaita, Andrea L.; Spinelli, Alessandro Sottocornola; Compagnoni, Christian Monzio

被引用次数: 14

2019

A Novel Single-Inductor Injection-Locked Frequency Divider by Three With Dual-Injection Secondary Locking

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Garghetti, Alessandro; Lacaita, Andrea L.; Levantino, Salvatore

被引用次数: 11

2019

A 30-GHz Digital Sub-Sampling Fractional-N PLL With-238.6-dB Jitter-Power Figure of Merit in 65-nm LP CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Bertulessi, Luca; Karman, Saleh; Cherniak, Dmytro; Garghetti, Alessandro; Samori, Carlo; Lacaita, Andrea L.; Levantino, Salvatore

被引用次数: 14

2019

A 1.6-to-3.0-GHz Fractional- N MDLL With a Digital-to-Time Converter Range-Reduction Technique Achieving 397-fs Jitter at 2.5-mW Power

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Santiccioli, Alessio; Mercandelli, Mario; Lacaita, Andrea L.; Samori, Carlo; Levantino, Salvatore

被引用次数: 30

2019

Efficient Behavioral Simulation of Charge-Pump Phase-Locked Loops

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Leoncini, Mauro; Bonfanti, Andrea; Levantino, Salvatore; Lacaita, Andrea L.

被引用次数: 11

2018

Investigation and Compact Modeling of the Time Dynamics of the GIDL-Assisted Increase of the String Potential in 3-D NAND Flash Arrays

IEEE TRANSACTIONS ON ELECTRON DEVICES

Malavena, Gerardo; Lacaita, Andrea L.; Spinelli, Alessandro S.; Compagnoni, Christian Monzio

被引用次数: 27

2018

Characterization and Modeling of Temperature Effects in 3-D NAND Flash Arrays-Part II: Random Telegraph Noise

IEEE TRANSACTIONS ON ELECTRON DEVICES

Nicosia, Gianluca; Mannara, Aurelio; Resnati, Davide; Paolucci, Giovanni M.; Tessariol, Paolo; Spinelli, Alessandro S.; Lacaita, Andrea L.; Goda, Akira; Compagnoni, Christian Monzio

被引用次数: 26

2018

Characterization and Modeling of Temperature Effects in 3-D NAND Flash Arrays-Part I: Polysilicon-Induced Variability

IEEE TRANSACTIONS ON ELECTRON DEVICES

Resnati, Davide; Mannara, Aurelio; Nicosia, Gianluca; Paolucci, Giovanni M.; Tessariol, Paolo; Spinelli, Alessandro S.; Lacaita, Andrea L.; Compagnoni, Christian Monzio

被引用次数: 53

2018

A Background Calibration Technique to Control the Bandwidth of Digital PLLs

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Mercandelli, Mario; Grimaldi, Luigi; Bertulesi, Luca; Samori, Carlo; Lacaita, Andrea L.; Levantino, Salvatore

被引用次数: 18

2018

Reviewing the Evolution of the NAND Flash Technology

PROCEEDINGS OF THE IEEE

Compagnoni, Christian Monzio; Goda, Akira; Spinelli, Alessandro S.; Feeley, Peter; Lacaita, Andrea L.; Visconti, Angelo

被引用次数: 176

2017

Investigation of the Program Operation of NAND Flash Cells With a Single-Electron Resolution

IEEE TRANSACTIONS ON ELECTRON DEVICES

Nicosia, Gianluca; Compagnoni, Christian Monzio; Paolucci, Giovanni M.; Resnati, Davide; Miccoli, Carmine; Spinelli, Alessandro S.; Lacaita, Andrea L.; Visconti, Angelo; Goda, Akira

被引用次数: 2

2016

A 64-Channel 965- μ W Neural Recording SoC With UWB Wireless Transmission in 130-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Brenna, Stefano; Padovan, Fabio; Neviani, Andrea; Bevilacqua, Andrea; Bonfanti, Andrea; Lacaita, Andrea L.

被引用次数: 21

2016

Random Telegraph Noise-Induced Sensitivity of Data Retention to Cell Position in the Programmed Distribution of NAND Flash Memory Arrays

IEEE ELECTRON DEVICE LETTERS

Resnati, Davide; Compagnoni, Christian Monzio; Paolucci, Giovanni M.; Miccoli, Carmine; Spinelli, Alessandro S.; Lacaita, Andrea L.; Visconti, Angelo; Goda, Akira

被引用次数: 4

2015

A Sub-400-nT/ $\sqrt{\text{Hz}}$, 775- μ W, Multi-Loop MEMS Magnetometer With Integrated Readout Electronics

JOURNAL OF MICROELECTROMECHANICAL SYSTEMS

Minotti, Paolo; Brenna, Stefano; Laghi, Giacomo; Bonfanti, Andrea G.; Langfelder, Giacomo; Lacaita, Andrea L.

被引用次数: 23

2015

First Detection of Single-Electron Charging of the Floating Gate of NAND Flash Memory Cells

IEEE ELECTRON DEVICE LETTERS

Compagnoni, Christian Monzio; Paolucci, Giovanni M.; Miccoli, Carmine; Spinelli, Alessandro S.; Lacaita, Andrea L.; Visconti, Angelo; Goda, Akira

被引用次数: 8

2015

Investigation of the Turn-ON of T-RAM Cells Under Transient Conditions

IEEE TRANSACTIONS ON ELECTRON DEVICES

Mulaosmanovic, Halid; Compagnoni, Christian Monzio; Castellani, Niccolo; Paolucci, Giovanni M.; Carnevale, Gianpietro; Fantini, Paolo; Ventrice, Domenico; Lacaita, Andrea L.; Spinelli, Alessandro S.; Benvenuti, Augusto

被引用次数: 11

2015

A 1.7 GHz Fractional-N Frequency Synthesizer Based on a Multiplying Delay-Locked Loop

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Levantino, Salvatore; Marucci, Giovanni; Marzin, Giovanni; Fenaroli, Andrea; Samori, Carlo; Lacaita, Andrea L.

被引用次数: 59

2015

Modeling of Dynamic Operation of T-RAM Cells

IEEE TRANSACTIONS ON ELECTRON DEVICES

Resnati, Davide; Compagnoni, Christian Monzio; Mulaosmanovic, Halid; Castellani, Niccolo; Carnevale, Gianpietro; Fantini, Paolo; Ventrice, Domenico; Lacaita, Andrea L.; Spinelli, Alessandro S.; Benvenuti, Augusto

被引用次数: 10

2015

Fitting Cells Into a Narrow VT Interval: Physical Constraints Along the Lifetime of an Extremely Scaled NAND Flash Memory Array

IEEE TRANSACTIONS ON ELECTRON DEVICES

Paolucci, Giovanni Maria; Compagnoni, Christian Monzio; Spinelli, Alessandro S.; Lacaita, Andrea L.; Goda, Akira

被引用次数: 14

2015

Revisiting Charge Trapping/Detrapping in Flash Memories From a Discrete and Statistical Standpoint-Part I: VT Instabilities

IEEE TRANSACTIONS ON ELECTRON DEVICES

Paolucci, Giovanni M.; Compagnoni, Christian Monzio; Miccoli, Carmine; Spinelli, Alessandro S.; Lacaita, Andrea L.; Visconti, Angelo

被引用次数: 21

2014

Revisiting Charge Trapping/Detrapping in Flash Memories From a Discrete and Statistical Standpoint-Part II: On-Field Operation and Distributed-Cycling Effects

IEEE TRANSACTIONS ON ELECTRON DEVICES

Paolucci, Giovanni M.; Compagnoni, Christian Monzio; Miccoli, Carmine; Spinelli, Alessandro S.; Lacaita, Andrea L.; Visconti, Angelo

被引用次数: 18

2014

Electrical Conductivity Discontinuity at Melt in Phase Change Memory

IEEE ELECTRON DEVICE LETTERS

Crespi, Luca; Ghetti, Andrea; Boniardi, Mattia; Lacaita, Andrea L.

被引用次数: 20

2014

Working Principles of a DRAM Cell Based on Gated-Thyristor Bistability

IEEE ELECTRON DEVICE LETTERS

Mulaosmanovic, Halid; Paolucci, Giovanni M.; Compagnoni, Christian Monzio; Castellani, Niccolo; Carnevale, Gianpietro; Fantini, Paolo; Ventrice, Domenico; Lacaita, Andrea L.; Spinelli, Alessandro S.; Benvenuti, Augusto

被引用次数: 22

2014

Investigation of the RTN Distribution of Nanoscale MOS Devices From Subthreshold to On-State

IEEE ELECTRON DEVICE LETTERS

Amoroso, Salvatore M.; Compagnoni, Christian Monzio; Ghetti, Andrea; Gerrer, Louis; Spinelli, Alessandro S.; Lacaita, Andrea L.; Asenov, Asen

被引用次数: 35

2013

Suppression of Flicker Noise Up-Conversion in a 65-nm CMOS VCO in the 3.0-to-3.6 GHz Band

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Pepe, Federico; Bonfanti, Andrea; Levantino, Salvatore; Samori, Carlo; Lacaita, Andrea L.

被引用次数: 53

2013

A Wideband Fractional-N PLL With Suppressed Charge-Pump Noise and Automatic Loop Filter Calibration

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Levantino, Salvatore; Marzin, Giovanni; Samori, Carlo; Lacaita, Andrea L.

被引用次数: 49

2013

Analysis and Minimization of Flicker Noise Up-Conversion in Voltage-Biased Oscillators

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Pepe, Federico; Bonfanti, Andrea; Levantino, Salvatore; Samori, Carlo; Lacaita, Andrea L.

被引用次数: 36

2013

Investigation of the RTN amplitude statistics of nanoscale MOS devices by the statistical impedance field method

JOURNAL OF COMPUTATIONAL ELECTRONICS

Torrente, Giulio; Castellani, Niccolo; Ghetti, Andrea; Compagnoni, Christian Monzio; Lacaita, Andrea L.; Spinelli, Alessandro S.; Benvenuti, Augusto

被引用次数: 0

2013

Accuracy and Issues of the Spectroscopic Analysis of RTN Traps in Nanoscale MOSFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Adamu-Lema, Fikru; Compagnoni, Christian Monzio; Amoroso, Salvatore M.; Castellani, Niccolo; Gerrer, Louis; Markov, Stanislav; Spinelli, Alessandro S.; Lacaita, Andrea L.; Asenov, Asen

被引用次数: 25

2013

The race of phase change memories to nanoscale storage and applications

MICROELECTRONIC ENGINEERING

Lacaita, Andrea L.; Redaelli, Andrea

被引用次数: 78

2013

Characterization and Modeling of the Band-to-Band Current Variability of Nanoscale Device Junctions

IEEE TRANSACTIONS ON ELECTRON DEVICES

Ghetti, Andrea; Compagnoni, Christian Monzio; Calloni, Andrea; Vendrame, Loris; Spinelli, Alessandro S.; Lacaita, Andrea L.

被引用次数: 3

2013

New Erase Constraint for the Junction-Less Charge-Trap Memory Array in Cylindrical Geometry

IEEE TRANSACTIONS ON ELECTRON DEVICES

Maconi, Alessandro; Compagnoni, Christian Monzio; Spinelli, Alessandro S.; Lacaita, Andrea L.

被引用次数: 2

2013

Reliability Characterization Issues for Nanoscale Flash Memories: A Case Study on 45-nm NOR Devices

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Miccoli, Carmine; Compagnoni, Christian Monzio; Chiavarone, Luca; Beltrami, Silvia; Lacaita, Andrea L.; Spinelli, Alessandro S.; Visconti, Angelo

被引用次数: 4

2013

Study of Cycling-Induced Parameter Variations in Phase Change Memory Cells

IEEE ELECTRON DEVICE LETTERS

Boniardi, Mattia; Redaelli, Andrea; Ghetti, Andrea; Lacaita, Andrea L.

被引用次数: 7

2013

Dynamic Analysis of Current-Voltage Characteristics of Nanoscale Gated-Thyristors

IEEE ELECTRON DEVICE LETTERS

Paolucci, Giovanni M.; Compagnoni, Christian Monzio; Castellani, Niccolo; Carnevale, Gianpietro; Fantini, Paolo; Ventrice, Domenico; Lacaita, Andrea L.; Spinelli, Alessandro S.; Benvenuti, Augusto

被引用次数: 13

2013

String Current in Decananometer NAND Flash Arrays: A Compact-Modeling Investigation

IEEE TRANSACTIONS ON ELECTRON DEVICES

Paolucci, Giovanni M.; Miccoli, Carmine; Compagnoni, Christian Monzio; Spinelli, Alessandro S.; Lacaita, Andrea L.

被引用次数: 7

2012

Three-Dimensional Electrostatics- and Atomistic Doping-Induced Variability of RTN Time Constants in Nanoscale MOS Devices-Part I: Physical Investigation

IEEE TRANSACTIONS ON ELECTRON DEVICES

Castellani, Niccolo; Compagnoni, Christian Monzio; Mauri, Aurelio; Spinelli, Alessandro S.; Lacaita, Andrea L.

被引用次数: 15

2012

Three-Dimensional Electrostatics- and Atomistic Doping-Induced Variability of RTN Time Constants in Nanoscale MOS Devices-Part II: Spectroscopic Implications

IEEE TRANSACTIONS ON ELECTRON DEVICES

Compagnoni, Christian Monzio; Castellani, Niccolo; Mauri, Aurelio; Spinelli, Alessandro S.; Lacaita, Andrea L.

被引用次数: 19

2012

Quantum-Mechanical Charge Distribution in Cylindrical Gate-All-Around MOS Devices

IEEE TRANSACTIONS ON ELECTRON DEVICES

Spinelli, Alessandro S.; Compagnoni, Christian Monzio; Maconi, Alessandro; Amoroso, Salvatore Maria; Lacaita, Andrea L.

被引用次数: 7

2012

Efficient Calculation of the Impulse Sensitivity Function in Oscillators

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Levantino, Salvatore; Maffezzoni, Paolo; Pepe, Federico; Bonfanti, Andrea; Samori, Carlo; Lacaita, Andrea L.

被引用次数: 29

2012

Flicker Noise Up-Conversion due to Harmonic Distortion in Van der Pol CMOS Oscillators

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Bonfanti, Andrea; Pepe, Federico; Samori, Carlo; Lacaita, Andrea L.

被引用次数: 46

2012

A glitch-corrector circuit for low-spur ADPLLs

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Zanuso, Marco; Levantino, Salvatore; Samori, Carlo; Lacaita, Andrea L.

被引用次数: 0

2012

A 20 Mb/s Phase Modulator Based on a 3.6 GHz Digital PLL With-36 dB EVM at 5 mW Power

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Marzin, Giovanni; Levantino, Salvatore; Samori, Carlo; Lacaita, Andrea L.

被引用次数: 84

2012

Semi-Analytical Model for the Transient Operation of Gate-All-Around Charge-Trap Memories

IEEE TRANSACTIONS ON ELECTRON DEVICES

Amoroso, Salvatore Maria; Compagnoni, Christian Monzio; Mauri, Aurelio; Maconi, Alessandro; Spinelli, Alessandro S.; Lacaita, Andrea L.

被引用次数: 37

2011

Three-Dimensional Simulation of Charge-Trap Memory Programming-Part I: Average Behavior

IEEE TRANSACTIONS ON ELECTRON DEVICES

Amoroso, Salvatore Maria; Maconi, Alessandro; Mauri, Aurelio; Compagnoni, Christian Monzio; Spinelli, Alessandro S.; Lacaita, Andrea L.

被引用次数: 14

2011

Three-Dimensional Simulation of Charge-Trap Memory Programming-Part II: Variability

IEEE TRANSACTIONS ON ELECTRON DEVICES

Maconi, Alessandro; Amoroso, Salvatore Maria; Compagnoni, Christian Monzio; Mauri, Aurelio; Spinelli, Alessandro S.; Lacaita, Andrea L.

被引用次数: 4

2011

Phase change materials in non-volatile storage

MATERIALS TODAY

Ielmini, Daniele; Lacaita, Andrea L.

被引用次数: 154

2011

Impact of Ge-Sb-Te compound engineering on the set operation performance in phase-change memories

SOLID-STATE ELECTRONICS

Boniardi, Mattia; Ielmini, Daniele; Tortorelli, Innocenzo; Redaelli, Andrea; Pirovano, Agostino; Allegra, Mario; Magistretti, Michele; Bresolin, Camillo; Erbetta, Davide; Modelli, Alberto; Varesi, Enrico; Pellizzer, Fabio; Lacaita, Andrea L.; Bez, Roberto

被引用次数: 23

2011

A 2.9-4.0-GHz Fractional-N Digital PLL With Bang-Bang Phase Detector and 560-fsrms Integrated Jitter at 4.5-mW Power

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Tasca, Davide; Zanusso, Marco; Marzin, Giovanni; Levantino, Salvatore; Samori, Carlo; Lacaita, Andrea L.

被引用次数: 196

2011

A Wideband 3.6 GHz Digital $\Delta\Sigma$ Fractional-N PLL With Phase Interpolation Divider and Digital Spur Cancellation

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zanusso, Marzo; Levantino, Salvatore; Samori, Carlo; Lacaita, Andrea L.

被引用次数: 91

2011

Low-Power Divider Retiming in a 3-4 GHz Fractional-N PLL

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Tasca, Davide; Zanusso, Marco; Levantino, Salvatore; Samori, Carlo; Lacaita, Andrea L.

被引用次数: 11

2011

Incomplete Filament Crystallization During Set Operation in PCM Cells

IEEE ELECTRON DEVICE LETTERS

Mantegazza, Davide; Ielmini, Daniele; Pirovano, Agostino; Lacaita, Andrea L.

被引用次数: 14

2010

Transient Simulation of Delay and Switching Effects in Phase-Change Memories

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lavizzari, Simone; Ielmini, Daniele; Lacaita, Andrea L.

被引用次数: 23

2010

Statistics of Resistance Drift Due to Structural Relaxation in Phase-Change Memory Arrays

IEEE TRANSACTIONS ON ELECTRON DEVICES

Boniardi, Mattia; Ielmini, Daniele; Lavizzari, Simone; Lacaita, Andrea L.; Redaelli, Andrea; Pirovano, Agostino

被引用次数: 82

2010

Comprehensive Investigation of Statistical Effects in Nitride Memories-Part I: Physics-Based Modeling

IEEE TRANSACTIONS ON ELECTRON DEVICES

Mauri, Aurelio; Compagnoni, Christian Monzio; Amoroso, Salvatore Maria; Maconi, Alessandro; Ghetti, Andrea; Spinelli, Alessandro S.; Lacaita, Andrea L.

被引用次数: 28

2010

Comprehensive Investigation of Statistical Effects in Nitride Memories-Part II: Scaling Analysis and Impact on Device Performance

IEEE TRANSACTIONS ON ELECTRON DEVICES

Compagnoni, Christian Monzio; Mauri, Aurelio; Amoroso, Salvatore Maria; Maconi, Alessandro; Greco, Eugenio; Spinelli, Alessandro S.; Lacaita, Andrea L.

被引用次数: 14

2010

Time-to-Digital Converter for Frequency Synthesis Based on a Digital Bang-Bang DLL

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Zanuso, Marco; Madoglio, Paolo; Levantino, Salvatore; Samori, Carlo; Lacaita, Andrea L.

被引用次数: 53

2010

Folding of Phase Noise Spectra in Charge-Pump Phase-Locked Loops Induced by Frequency Division

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Levantino, Salvatore; Collamati, Luca; Samori, Carlo; Lacaita, Andrea L.

被引用次数: 20

2010

Impact of Control-Gate and Floating-Gate Design on the Electron-Injection Spread of Decananometer NAND Flash Memories

IEEE ELECTRON DEVICE LETTERS

Compagnoni, Christian Monzio; Miccoli, Carmine; Lacaita, Andrea L.; Marmiroli, Andrea; Spinelli, Alessandro S.; Visconti, Angelo

被引用次数: 12

2010

Random Telegraph Noise Effect on the Programmed Threshold-Voltage Distribution of Flash Memories

IEEE ELECTRON DEVICE LETTERS

Compagnoni, Christian Monzio; Ghidotti, Michele; Lacaita, Andrea L.; Spinelli, Alessandro S.; Visconti, Angelo

被引用次数: 81

2009

Noise Analysis and Minimization in Bang-Bang Digital PLLs

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Zanuso, Marco; Tasca, Davide; Levantino, Salvatore; Donadel, Andrea; Samori, Carlo; Lacaita, Andrea L.

被引用次数: 90

2009

Reliability Impact of Chalcogenide-Structure Relaxation in Phase-Change Memory (PCM) Cells-Part II: Physics-Based Modeling

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lavizzari, Simone; Ielmini, Daniele; Sharma, Deepak; Lacaita, Andrea L.

被引用次数: 67

2009

Reliability Impact of Chalcogenide-Structure Relaxation in Phase-Change Memory (PCM) Cells-Part I: Experimental Study

IEEE TRANSACTIONS ON ELECTRON DEVICES

Ielmini, Daniele; Sharma, Deepak; Lavizzari, Simone; Lacaita, Andrea L.

被引用次数: 149

2009

Study of Multilevel Programming in Programmable Metallization Cell (PMC) Memory

IEEE TRANSACTIONS ON ELECTRON DEVICES

Russo, Ugo; Kamalanathan, Deepak; Ielmini, Daniele; Lacaita, Andrea L.; Kozicki, Michael N.

被引用次数: 290

2009

Statistical model for random telegraph noise in Flash memories

IEEE TRANSACTIONS ON ELECTRON DEVICES

Compagnoni, Christian Monzio; Gustneroli, Riccardo; Spinelli, Alessandro S.; Lacaita, Andrea L.; Bonanomi, Mauro; Visconti, Angelo

被引用次数: 101

2008

Modeling of programming and read performance in phase-change memories - Part II: Program disturb and mixed-scaling approach

IEEE TRANSACTIONS ON ELECTRON DEVICES

Russo, Ugo; Ielmini, Daniele; Redaelli, Andrea; Lacaita, Andrea L.

被引用次数: 44

2008

Modeling of programming and read performance in phase-change memories - Part I: Cell optimization and scaling

IEEE TRANSACTIONS ON ELECTRON DEVICES

Russo, Ugo; Ielmini, Daniele; Redaelli, Andrea; Lacaita, Andrea L.

被引用次数: 106

2008

Low-power signal component separator for a 64-QAM 802.11 LINC transmitter

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Panseri, Luigi; Romano, Luca; Levantino, Salvatore; Samori, Carlo; Lacaita, Andrea L.

被引用次数: 10

2008

Phase-change memories

PHYSICA STATUS SOLIDI A-APPLICATIONS AND MATERIALS SCIENCE

Lacaita, Andrea L.; Wouters, Dirk J.

被引用次数: 127

2008

Voltage-controlled relaxation oscillations in phase-change memory devices

IEEE ELECTRON DEVICE LETTERS

Ielmini, Daniele; Mantegazza, Davide; Lacaita, Andrea L.

被引用次数: 29

2008

Quantization effects in all-digital phase-locked loops

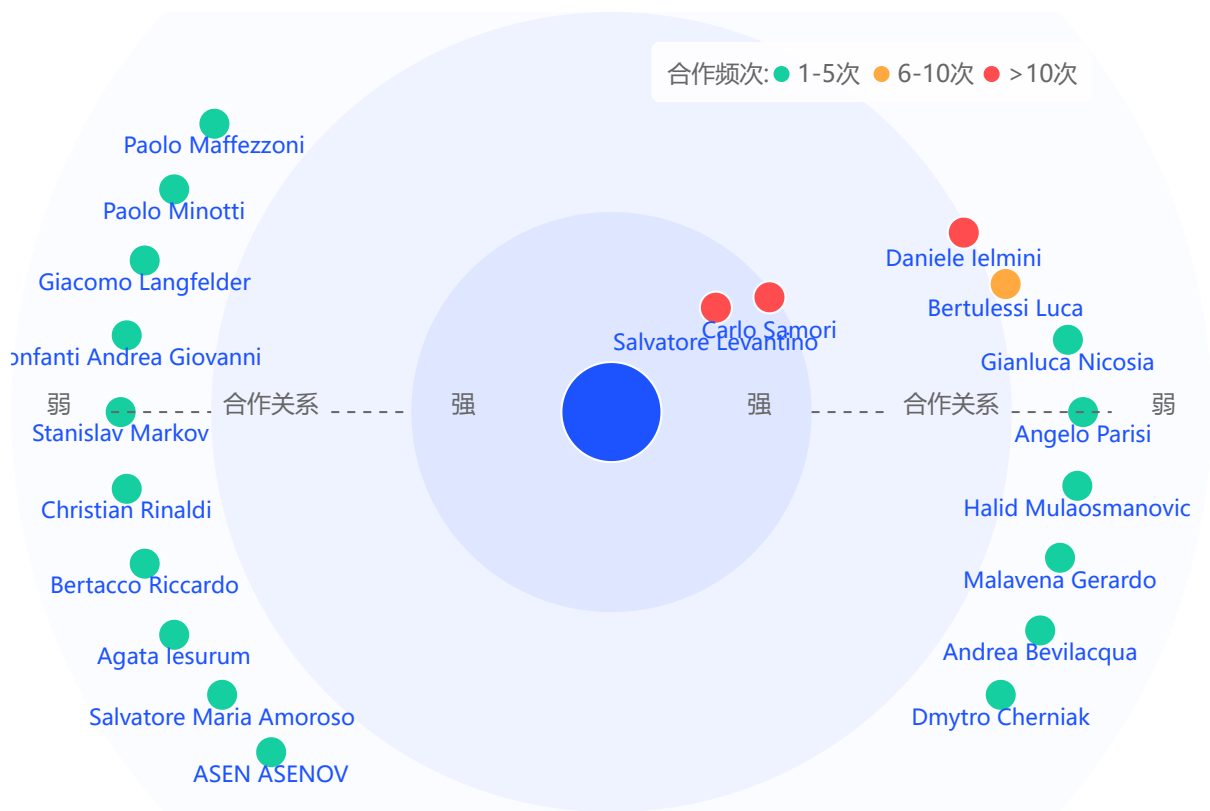
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Madoglio, Paolo; Zanuso, Marco; Levantino, Salvatore; Samori, Carlo; Lacaita, Andrea L.

被引用次数: 38

2007

学术合作



Salvatore Levantino

米兰理工大学

合作次数: 37

Carlo Samori

米兰理工大学

合作次数: 32

Daniele Ielmini

米兰理工大学

合作次数: 11

Bertulesi Luca

米兰理工大学

合作次数: 9

Gianluca Nicosia

米兰理工大学

合作次数: 4

还有 15 个合作者